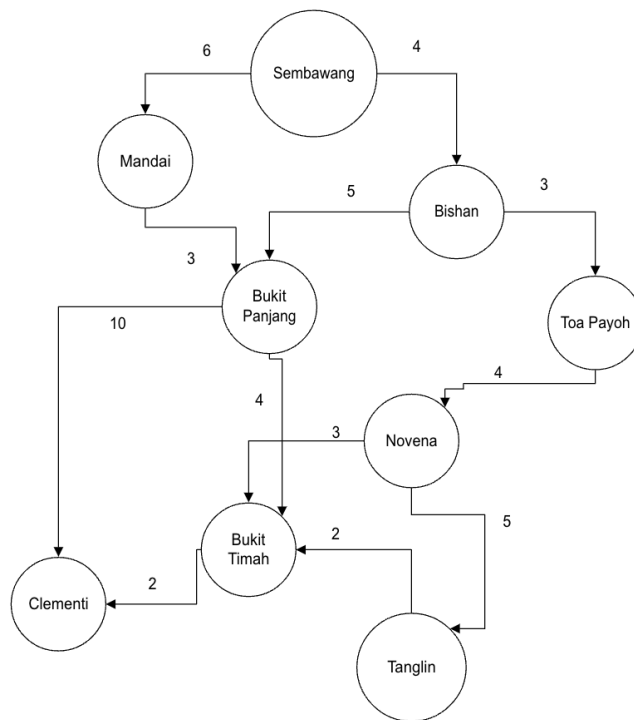


IT5005 Artificial Intelligence
Final Exam
SOLUTIONS

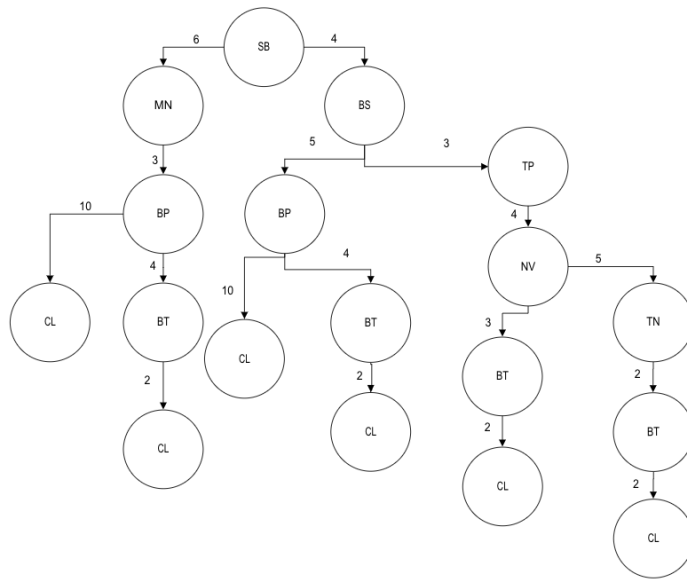
QUESTION 1. SEARCH TREES (25 MARKS)

We are given the following map of Singapore. The number shown on the edges are the distances between the towns in kilometers. Our friend Aiken would like travel from Sembawang to Clementi.



From the map above Aiken managed to draw out this tree, shortening the names to just two letters per town:

SB: Sembawang, MN: Mandai, BS: Bishan, BP: Bukit Panjang, BT: Bukit Timah, CL: Clementi, TP: Toa Payoh, NV: Novena, TN: Tanglin



- a. If we performed breadth first search (BFS), what is the sequence of towns visited? (8 marks)

Eight blanks are provided. Skip Sembawang (SB) (since all routes start from Sembawang), and fill one town name per blank using the two-letter abbreviations. Fill "UN" for unused boxes.

Recall: Aiken wants to get from Sembawang to Clementi.

For example if the towns visited by the BFS algorithm are Sembawang, Bishan, Bukit Timah and Clementi, fill in:

First blank: BS

Second blank: BT

Third blank: CL

Fourth to eighth blanks: UN

SB

MN BS

BS BP

BP TP

TP CL BT

CL BT NV

Arrived

Path SB->MN->BP1->CL1

- b. What is the total distance from Sembawang to Clementi with breadth first search? Fill in only numbers. E.g. if the answer is 72 kilometers, just fill in 72. (2 marks)

Sembawang->Mandai->Bukit Panjang->Clementi

Distance = 6+3+10 = 19 km

- c. If we performed depth first search, which of the following shows the sequence of towns visited? (8 marks)

Eight blanks are provided. Skip Sembawang (SB) (since all routes start from Sembawang), and fill one town name per blank using the two-letter abbreviations. Fill "UN" for unused boxes.

Recall: Aiken wants to get from Sembawang to Clementi.

For example if the towns visited are Sembawang, Bishan, Bukit Timah and Clementi, fill in:

First blank: BS

Second blank: BT

Third blank: CL

Fourth to eighth blanks: UN

SB

MN BS

BP BS

CL BT

Reached

- d. What is the total distance from Sembawang to Clementi with depth first search? Fill in only numbers. E.g. if the answer is 72 kilometers, just fill in 72. (2 marks)

Sembawang, Mandai, Bukit Panjang, Clementi

19 km

- e. What is the route and total distance from Sembawang to Clementi under Dijkstra's Algorithm? You may use either the original map or the tree version to solve this question.

Eight blanks are provided. Skip Sembawang (SB) (since all routes start from Sembawang), and fill one town name per blank using the two-letter abbreviations. Fill "UN" for unused boxes. (5 marks)

Recall: Aiken wants to get from Sembawang to Clementi.

We are asking for **the route from Sembawang to Clementi**, NOT the nodes visited by the algorithm.

For example if **the route** is Sembawang to Bishan to Bukit Timah to Clementi, fill up the blanks like this:

First blank: BS

Second blank: BT

Third blank: CL

Fourth to eighth blanks: UN

For the distance, Fill in only numbers. E.g. if the answer is 72 kilometers, just fill in 72.

If you write your answers in any other format it will be marked as incorrect.

Step	V(T)	E(T)	F	L(SB)	L(MN)	L(BS)	L(BP)	L(BT)	L(TP)	L(NV)	L(TN)	L(CL)
0	{SB}	-	{SB}	0	inf	inf	inf	inf	inf	inf	inf	inf
1	{SB}	-	{MN, <u>BS</u> }	0	6	<u>4</u>	inf	inf	inf	inf	inf	inf
2	{SB, BS}	{[SB, BS]}	{ <u>MN</u> , BP, TP}	0	<u>6</u>	<u>4</u>	4+5 = 9	Inf	4 + 3 = 7	Inf	Inf	Inf
3	{SB, BS, MN}	{[SB, BS], [SB, MN],	{BP, <u>TP</u> }	0	<u>6</u>	<u>4</u>	min(9, 9) = 9	Inf	<u>7</u>	Inf	Inf	Inf
4	{SB, BS, MN, TP}	{[SB, BS], [SB, MN], [BS, TP]}	{ <u>BP</u> , NV}	0	<u>6</u>	<u>4</u>	<u>9</u>	Inf	<u>7</u>	7 + 4 = 11	Inf	Inf
5	{SB, BS, MN, TP, BP}	{[SB, BS], [SB, MN], [BS, TP], [BS, BP]}	{ <u>NV</u> , BT, CL}	0	<u>6</u>	<u>4</u>	<u>9</u>	13	<u>7</u>	<u>11</u>	Inf	9 + 10 = 19
6	{SB, BS, MN, TP, BP, NV}	{[SB, BS], [SB, MN], [BS, TP], [BS, BP], [TP, NV]}	{TN, <u>BT</u> , CL}	0	<u>6</u>	<u>4</u>	<u>9</u>	<u>13</u>	<u>7</u>	<u>11</u>	11 + 5 = 16	19

Step	V(T)	E(T)	F	L(SB)	L(MN)	L(BS)	L(BP)	L(BT)	L(TP)	L(NV)	L(TN)	L(CL)
7	{SB, BS, MN, TP, BP, NV, BT}	{[SB, BS], [SB, MN], [BS, TP], [BS, BP], [TP, NV], [BP, BT]}	{TN, CL}	0	<u>6</u>	<u>4</u>	<u>9</u>	<u>13</u>	<u>7</u>	<u>11</u>	16	13 + 2 = <u>15</u>
8	{SB, BS, MN, TP, BP, NV, BT, CL}	{[SB, BS], [SB, MN], [BS, TP], [BS, BP], [TP, NV], [BP, BT], [BT, CL]}	{TN}	0	<u>6</u>	<u>4</u>	<u>9</u>	<u>13</u>	<u>7</u>	<u>11</u>	16	<u>15</u>

Route; SB->BS->BP->BT->CL = 4 + 5 + 4 + 2 = 15 km

QUESTION 2. UNSUPERVISED LEARNING (18 MARKS)

In this question we consider a 3 x 3 Kohonen map initialized as shown below. Each square has the format:

A (X, Y)

Where A is the unit number, and X and Y are elements of a 2-element vector, denoting the current weight values for that unit.

1 (1, 3)	2 (10,9)	3 (7, 12)
4 (8,2)	5 (17,5)	6 (21,8)
7 (9,21)	8 (14, 32)	9 (6,12)

We define the LR distance for two n element vectors X and Y given by:

$$L2(X, Y) = \sqrt{\sum_i^n (x_i - y_i)^2}$$

Complete the following table for the input vector U. Put Y in the BMU box if that unit is the BMU, otherwise put N.

Round your answers to the nearest 2 decimal places. For example, if your answer is 3.5, enter 3.50. If it is 2.485, enter 2.49. If it is 6.53. **If you write your answer in any other format you will be marked as incorrect.** (10 marks)

Input: U = (17, 32)

Unit	Current weight values	LR Distance	BMU?
1	(1,3)	$\sqrt{(17-1)^2 + (32-3)^2}$ $= \sqrt{1097}$ $= 33.12$	N
2	(10,9)	$\sqrt{\{(17-10)^2 + (32-9)^2\}}$ $= \sqrt{\{7^2 + 23^2\}}$ $= \sqrt{578}$ $= 24.04$	N
3	(7,12)	$\sqrt{(17-7)^2 + (32-12)^2} =$ $\sqrt{10^2 + 20^2} = \sqrt{500} =$ 22.36	N
4	(8, 2)	$= \sqrt{(17-8)^2 + (32-2)^2}$ $= \sqrt{9^2 + 30^2}$ $= \sqrt{981}$ $= 31.32$	N
5	(17, 5)	$\sqrt{(17-17)^2 + (32-5)^2}$ $= \sqrt{0^2 + 27^2}$ $= 27$	N
6	(21, 8)	$\sqrt{(17-21)^2 + (32-8)^2}$ $= \sqrt{-4^2 + 24^2} = \sqrt{592}$ $= 24.33$	N
7	(9, 21)	$\sqrt{(17-9)^2 + (32-21)^2}$ $= \sqrt{8^2 + 11^2}$ $= \sqrt{185}$ $= 13.60$	N
8	(14, 32)	$\sqrt{(17-14)^2 + (32-32)^2}$ $= \sqrt{3^2 + 0^2}$ $= 3.00$	Y
9	(6, 12)	$\sqrt{\{(17-6)^2 + (32-12)^2\}} = \sqrt{\{11^2 + 20^2\}} =$ $\sqrt{521} = 22.83$	N

QUESTION 3. CONSTRAINT SATISFACTION (15 MARKS)

- a. We are given the following constraint satisfaction problem (CSP) for 3 integer variables X, Y and Z, subject to the following constraints:

$$Y = 2X$$

$$Z = 5Y - 2$$

We are initially given the domains $D_X = D_Y = D_Z = \{0,1,2, \dots, 20\}$. Apply the AC3 algorithm to find suitable domains of X, Y and Z that are arc-consistent.

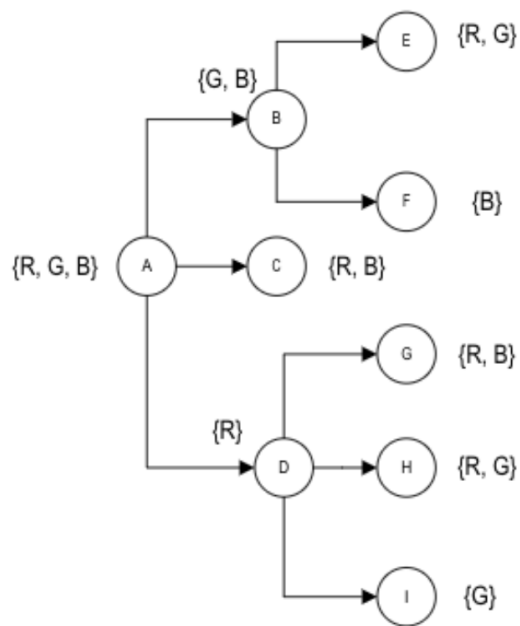
List the elements of the domains individually. For example if you find that $D_X = \{1..8\}$, write your answer as 1,2,3,4,5,6,7,8 without any spaces. Leave out the { and } braces. **If you write your answer in any other format you will be marked as incorrect.**

(6 marks)

Revised Domain	Queue
	(X, Y), (Y, X), (Y, Z), (Z, Y)
$D_x = \{0..10\}$	(Y, X), (Y, Z), (Z, Y)
$D_y = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$	(Y, Z), (Z, Y)
$D_y = \{2, 4\}$	(Z, Y), (X, Y)
$D_z = \{8, 18\}$	{X, Y}
$D_x = \{1, 2\}$	

Domains are $D_x = \{1, 2\}, D_y = \{2, 4\}, D_z = \{8, 18\}$

- b. We are given the following constraints graph, and our goal is to colour each node either red (R), green (G) or blue (B). Adjacent nodes (i.e. nodes with edges between them) must not have the same colour. Solve this constraint problem by filling in the colours for each node A to I. The set of possible colours is shown in the labels near each node.



Fill your answers as R for red, G for green and B for blue. **If you fill your answer in any other way it will be marked as incorrect.** (9 marks)

Colour for node A (R, G or B):

Colour for node B (R, G or B):

Colour for node C (R, G or B):

Colour for node D (R, G or B):

Colour for node E (R, G or B):

Colour for node F (R, G or B):

Colour for node G (R, G or B):

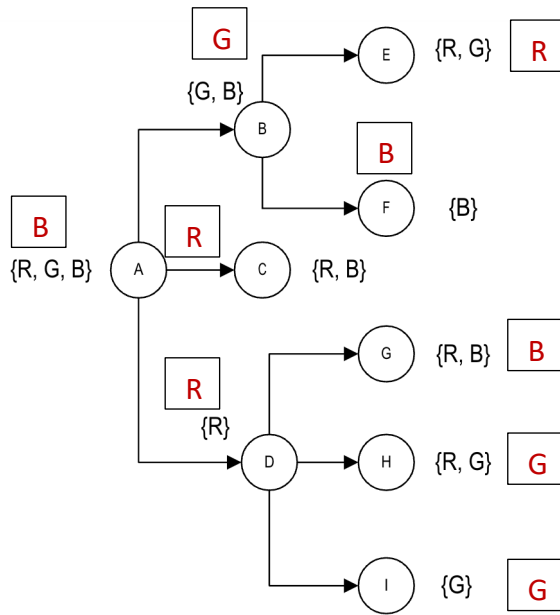
Colour for node H (R, G or B):

Colour for node I (R, G or B):

SOLUTION

G

R



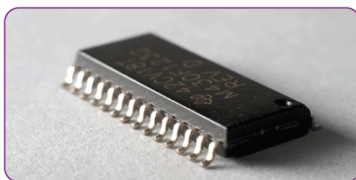
Answers:

Colour for node A (R, G or B): **B**
 Colour for node B (R, G or B): **G**
 Colour for node C (R, G or B): **R**
 Colour for node D (R, G or B): **R**
 Colour for node E (R, G or B): **R**
 Colour for node F (R, G or B): **B**
 Colour for node G (R, G or B): **B**
 Colour for node H (R, G or B): **G**
 Colour for node I (R, G or B): **G**

QUESTION 4. UNCERTAINTY (15 MARKS)

You have been given two large consignments of integrated circuits (ICs) to test. The picture below shows an example of an IC.

INTEGRATED CIRCUIT



ICs are either faulty, meaning that they don't work, or good, meaning that they work.

The first consignment is labelled "FAULTY", and the second is labelled "GOOD". For every 100 ICs in the FAULTY consignment, 95 will actually be faulty. For every 100 ICs in the GOOD consignment, 93 will actually be good.

You are given a test machine and your job is to test the ICs to see whether they are actually faulty or good.

Unfortunately your test machine itself is not 100% reliable. Out of every 1000 faulty ICs that are actually faulty, your machine will mis-identify 5 of them as good. Out of every 1000 ICs that are actually good, your machine will mis-identify 8 of them as faulty.

Find the following, expressing your answer rounded to the nearest 4 decimal places. For example, an answer of 0.95 should be entered as 0.9500, and answer of 0.94483 should be entered as 0.9448, and answer of 0.94438 should be entered as 0.9444, etc. **If you enter your answers in any other way, it will be marked as incorrect.**

- a. If you take an IC from the “FAULTY” consignment and the test machine identifies it as “FAULTY”, what is the probability that it is actually faulty? (5 marks)

Let F be the event that the IC is faulty and G be the event that the IC is good. Let DF be the event that the IC is detected as faulty, and DG be the event that the IC is detected as good.

We have:

$$\begin{aligned}P(DG|F) &= \frac{5}{1000} \\&= 0.005\end{aligned}$$

It thus detects 995 out of 1000 faulty ICs as faulty, i.e.

$$\begin{aligned}P(DF|F) &= 1 - P(DG|F) \\&= 0.995\end{aligned}$$

We also have:

$$\begin{aligned}P(DF|G) &= \frac{8}{1000} \\&= 0.008\end{aligned}$$

Thus:

$$\begin{aligned}P(DG|G) &= 1 - P(DF|G) \\&= 0.992\end{aligned}$$

Since the IC is drawn from the FAULTY consignment, $P(F) = 95/100 = 0.95$

$$\begin{aligned}P(F|DF) &= \frac{P(DF|F)P(F)}{P(DF|F)P(F) + P(DF|G)P(G)} \\&= \frac{0.995 * 0.95}{0.995 * 0.95 + 0.08 * 0.05}\end{aligned}$$

$$= \frac{0.94525}{0.94525 + 0.004} = 0.9958$$

- b. If you take an IC from the “GOOD” consignment and the test machine identifies it as “GOOD”, what is the probability that it is actually good? (5 marks)

Since the IC is drawn from the GOOD consignment, $P(G) = \frac{93}{100} = 0.93$

$$\begin{aligned} P(G|DG) &= \frac{P(DG|G)P(G)}{P(DG|G)P(G) + P(DG|F)P(F)} \\ &= \frac{0.992 * 0.93}{0.992 * 0.93 + 0.005 * 0.07} \\ &= \frac{0.92556}{0.92556 + 0.00035} \\ &= 0.9996 \end{aligned}$$

- c. If you take an IC from the “FAULTY” consignment and the test machine identifies it as “GOOD”, what is the probability that it is actually good? (5 marks)

Since we took from the FAULTY consignment, $P(F) = 0.95$, $P(G) = 0.05$

$$\begin{aligned} P(G|DG) &= \frac{P(DG|G)P(G)}{P(DG|G)P(G) + P(DG|F)P(F)} \\ &= \frac{0.992 * 0.05}{0.992 * 0.05 + 0.005 * 0.95} \\ &= \frac{0.0496}{0.0496 + 0.00475} \\ &= 0.9126 \end{aligned}$$